

HW4 – Implementing a FASTA file parser and calculating FASTA statistics using command line tools from Python

- 1) Create “HW4.ipynb” – this will be the file for your homework.
- 2) In this homework you will implement a FASTA parser that will read FASTA files, scan them for possible Uniprot/ENSEMBL IDs in definition line and calling corresponding databases for a detailed data.
- 3) Additionally, you will practice calling command line tool SeqKit from Python using subprocess module.
- 4) You have three fasta files. One is correct, one contains incorrect ID and one is invalid.
- 5) Your task is to write a parser that will do the following:
 - a. Call “seqkit stats” on a given fasta file.
 - b. Parse an output from the seqkit. If collection raised an error – capture it and return as a final output.
 - c. If the stats collection is successful, you should find a file type (DNA or Protein) in the output.
 - d. Read a given fasta file with biopython.
 - e. Iterate over sequences to read the descriptions.
 - f. While iterating, check each description for a possible ID from either Uniprot or ENSEMBL depending on what type you identified before (one fasta file contains only one type of data - nucleotides or aminoacids – so you can define your search criteria before starting an iteration) - use regular expressions from HW2_1.
 - g. Perform an API call to a given database (Uniprot or ENSEMBL) – use requests from HW2.
 - h. The final output should contain info about each sequence (description, sequence, info from DB and the name of this DB).
- 6) As for code structure, you have a template provided and test_data.txt to see the example. The parser can be implemented as a Class, containing
 - a. Several utility methods (starting with ___) for API calls and parsing responses – you can use your code from HW2.
 - b. Public methods that should be used for producing an actual output - seqkit call and biopython parser.
- 7) Test your work and save the output inside Jupyter notebook.
- 8) Create “HW4” branch in your repo.
- 9) Upload your ipynb file into “HW4” and create a pull request to submit your work.

Soft Deadline for this task is 25.03.2026 23.59 MSK. After this date, **half** points will be graded for the task.

Hard Deadline for this task is 1.04.2026 23.59 MSK. After this date, **no** points will be graded for the task.